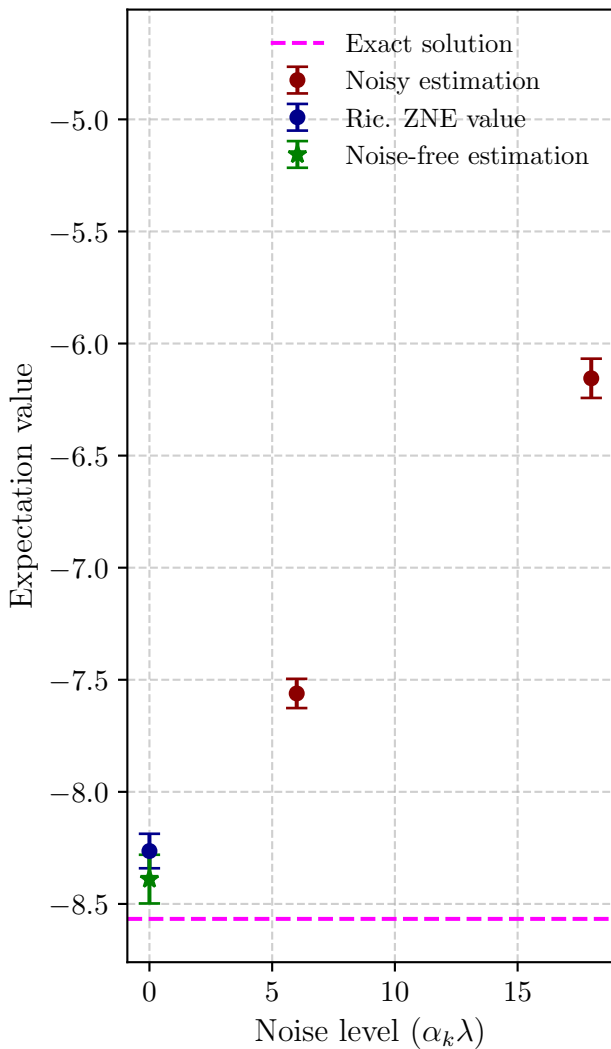
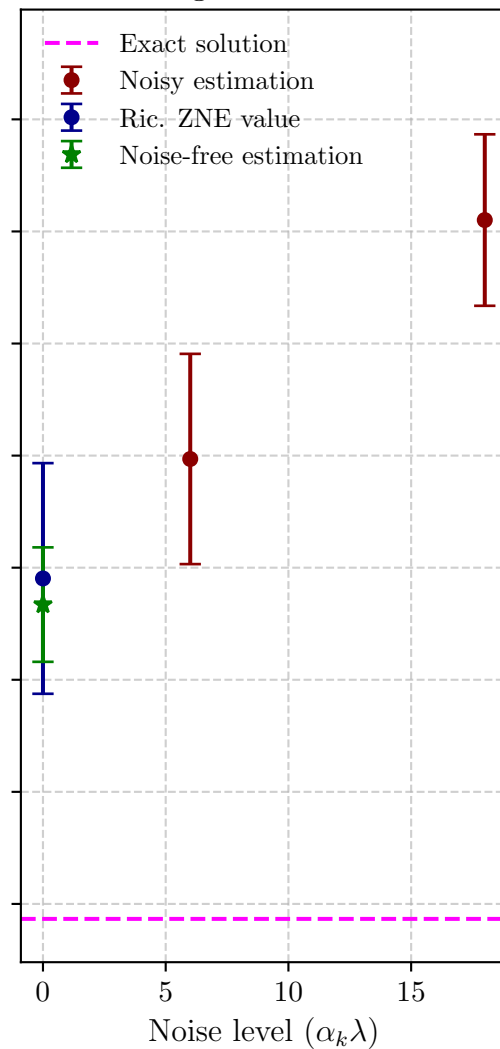


Description	Simulation	Mean $\pm$ Std
xy-ric2	vqe	-7.56107 $\pm$ 0.06520
xy-ric2	redundant (noise=6)	-7.56107 $\pm$ 0.06520
xy-ric2	redundant (noise=18)	-6.15518 $\pm$ 0.08758
xy-ric2	zne	-8.26402 $\pm$ 0.07690
xy-ric2	noiseoff	-8.38929 $\pm$ 0.10862
ising-ric2	vqe	-6.51507 $\pm$ 0.46899
ising-ric2	redundant (noise=6)	-6.51507 $\pm$ 0.46899
ising-ric2	redundant (noise=18)	-5.44914 $\pm$ 0.38250
ising-ric2	zne	-7.04804 $\pm$ 0.51449
ising-ric2	noiseoff	-7.16465 $\pm$ 0.25530
heisenberg-ric2	vqe	-6.26219 $\pm$ 0.17777
heisenberg-ric2	redundant (noise=6)	-6.26219 $\pm$ 0.17777
heisenberg-ric2	redundant (noise=18)	-4.92419 $\pm$ 0.22076
heisenberg-ric2	zne	-6.93119 $\pm$ 0.16633
heisenberg-ric2	noiseoff	-6.72634 $\pm$ 0.32239
xy-ric3-uniform-scaling	vqe	-7.56107 $\pm$ 0.06520
xy-ric3-uniform-scaling	redundant (noise=6)	-7.56107 $\pm$ 0.06520
xy-ric3-uniform-scaling	redundant (noise=18)	-6.15518 $\pm$ 0.08758
xy-ric3-uniform-scaling	redundant (noise=30)	-5.04735 $\pm$ 0.12295
xy-ric3-uniform-scaling	zne	-8.37580 $\pm$ 0.08182
xy-ric3-uniform-scaling	noiseoff	-8.38929 $\pm$ 0.10862
ising-ric3-uniform-scaling	vqe	-6.51507 $\pm$ 0.46899
ising-ric3-uniform-scaling	redundant (noise=6)	-6.51507 $\pm$ 0.46899
ising-ric3-uniform-scaling	redundant (noise=18)	-5.44914 $\pm$ 0.38250
ising-ric3-uniform-scaling	redundant (noise=30)	-4.56370 $\pm$ 0.32425
ising-ric3-uniform-scaling	zne	-7.11572 $\pm$ 0.52373
ising-ric3-uniform-scaling	noiseoff	-7.16465 $\pm$ 0.25530
heisenberg-ric3-uniform-scaling	vqe	-6.26219 $\pm$ 0.17777
heisenberg-ric3-uniform-scaling	redundant (noise=6)	-6.26219 $\pm$ 0.17777
heisenberg-ric3-uniform-scaling	redundant (noise=18)	-4.92419 $\pm$ 0.22076
heisenberg-ric3-uniform-scaling	redundant (noise=30)	-3.86627 $\pm$ 0.25204
heisenberg-ric3-uniform-scaling	zne	-7.03622 $\pm$ 0.16244
heisenberg-ric3-uniform-scaling	noiseoff	-6.72634 $\pm$ 0.32239
xy-ric4-uniform-scaling	vqe	-7.56107 $\pm$ 0.06520
xy-ric4-uniform-scaling	redundant (noise=6)	-7.56107 $\pm$ 0.06520
xy-ric4-uniform-scaling	redundant (noise=18)	-6.15518 $\pm$ 0.08758
xy-ric4-uniform-scaling	redundant (noise=30)	-5.04735 $\pm$ 0.12295
xy-ric4-uniform-scaling	redundant (noise=42)	-4.16757 $\pm$ 0.15007
xy-ric4-uniform-scaling	zne	-8.39767 $\pm$ 0.08306
xy-ric4-uniform-scaling	noiseoff	-8.38929 $\pm$ 0.10862
ising-ric4-uniform-scaling	vqe	-6.51507 $\pm$ 0.46899
ising-ric4-uniform-scaling	redundant (noise=6)	-6.51507 $\pm$ 0.46899
ising-ric4-uniform-scaling	redundant (noise=18)	-5.44914 $\pm$ 0.38250
ising-ric4-uniform-scaling	redundant (noise=30)	-4.56370 $\pm$ 0.32425
ising-ric4-uniform-scaling	redundant (noise=42)	-3.82308 $\pm$ 0.28550
ising-ric4-uniform-scaling	zne	-7.12687 $\pm$ 0.52568
ising-ric4-uniform-scaling	noiseoff	-7.16465 $\pm$ 0.25530
heisenberg-ric4-uniform-scaling	vqe	-6.26219 $\pm$ 0.17777
heisenberg-ric4-uniform-scaling	redundant (noise=6)	-6.26219 $\pm$ 0.17777
heisenberg-ric4-uniform-scaling	redundant (noise=18)	-4.92419 $\pm$ 0.22076
heisenberg-ric4-uniform-scaling	redundant (noise=30)	-3.86627 $\pm$ 0.25204
heisenberg-ric4-uniform-scaling	redundant (noise=42)	-3.02209 $\pm$ 0.26938
heisenberg-ric4-uniform-scaling	zne	-7.05696 $\pm$ 0.16129
heisenberg-ric4-uniform-scaling	noiseoff	-6.72634 $\pm$ 0.32239

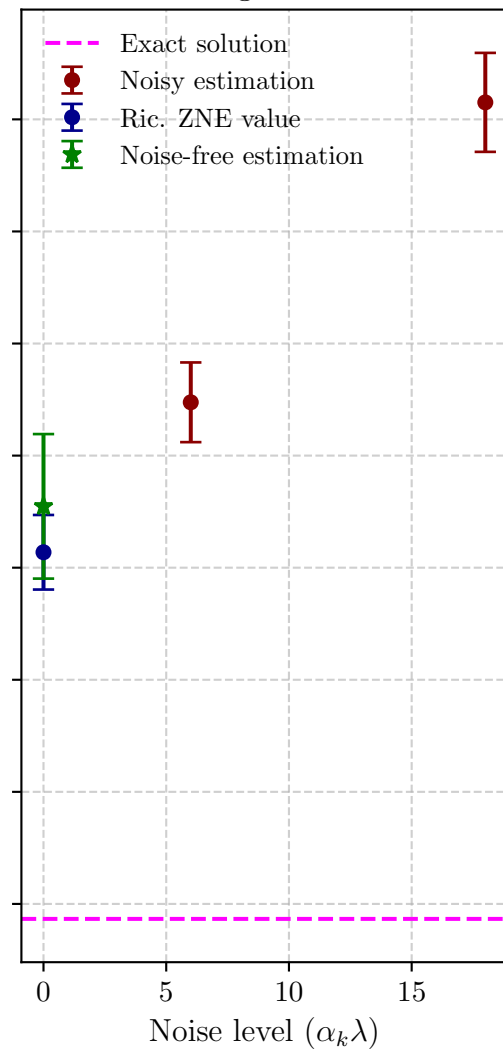
XY ansatz ric-2



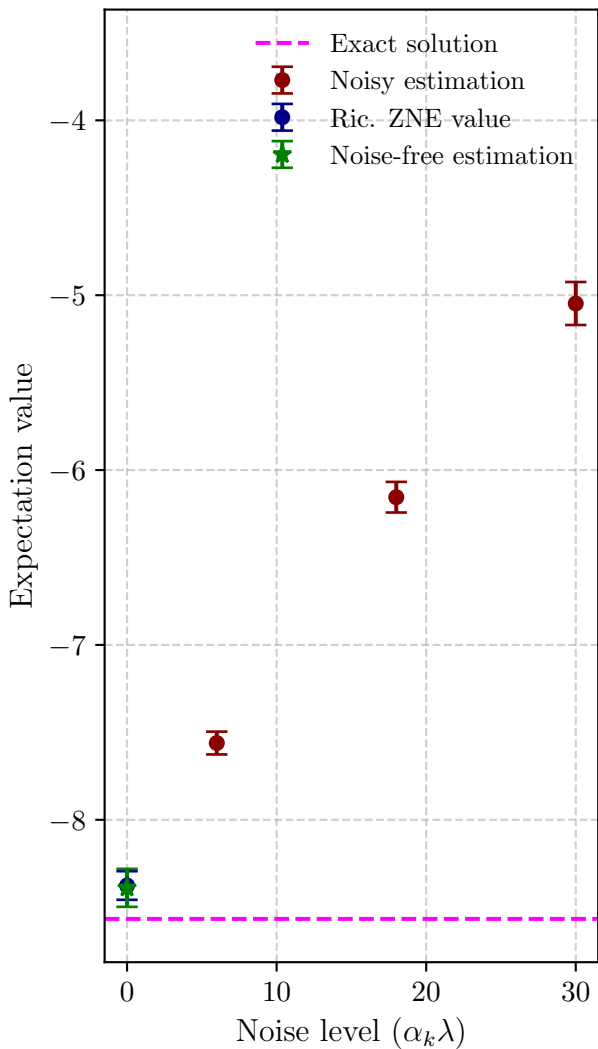
Ising ansatz ric-2



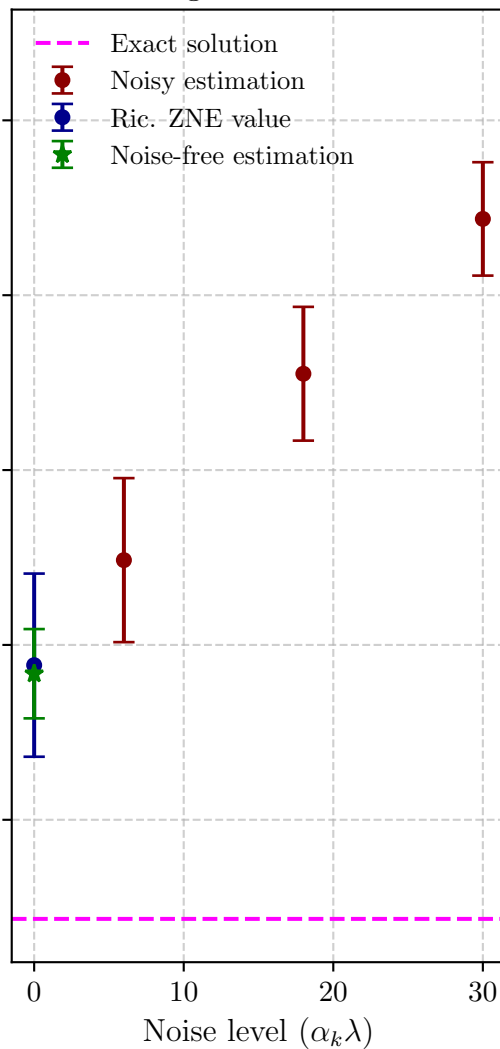
Heisenberg ansatz ric-2



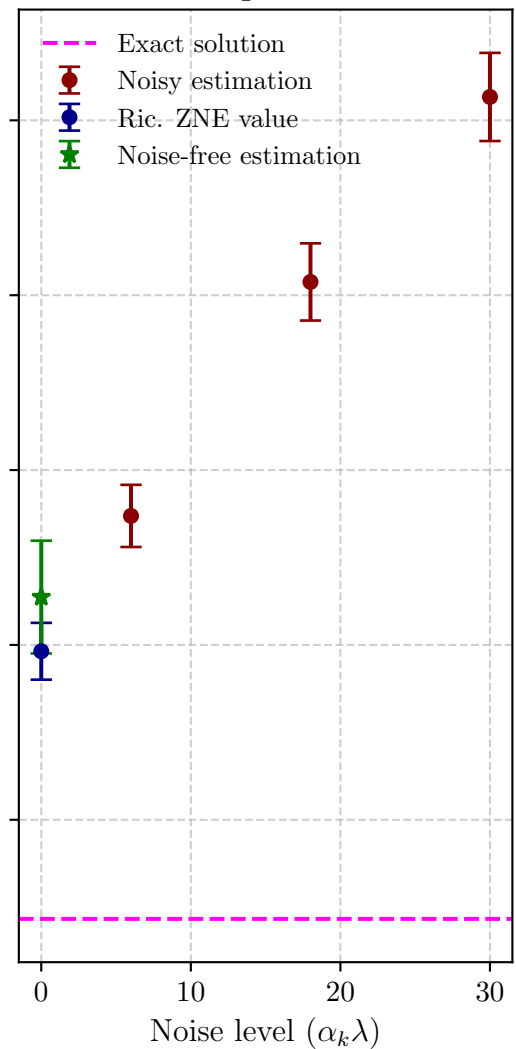
XY ansatz ric-3



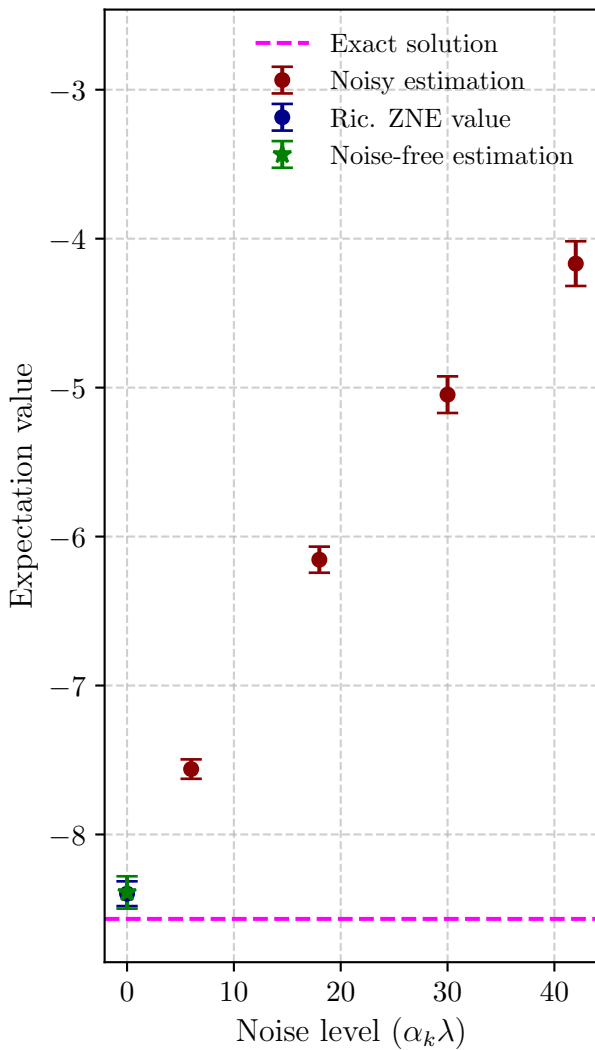
Ising ansatz ric-3



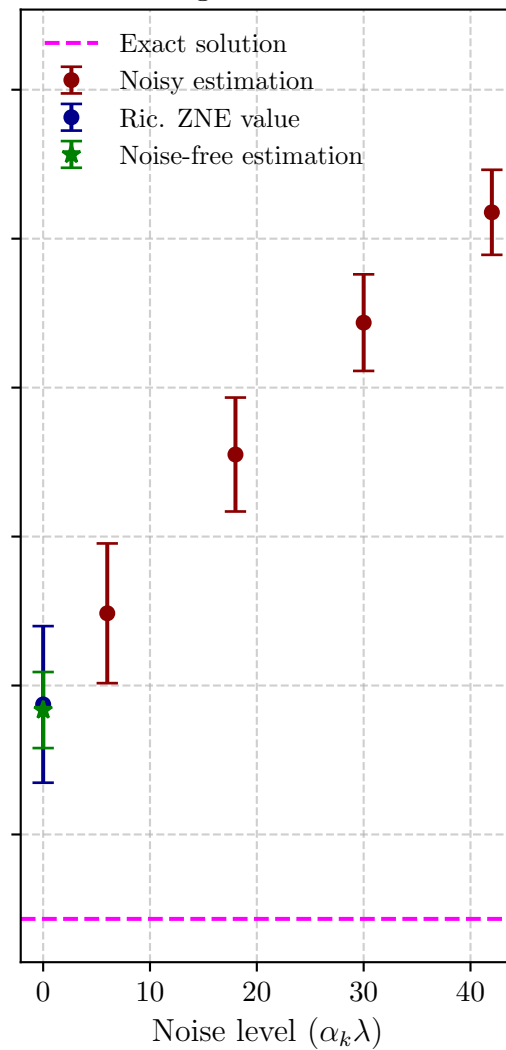
Heisenberg ansatz ric-3



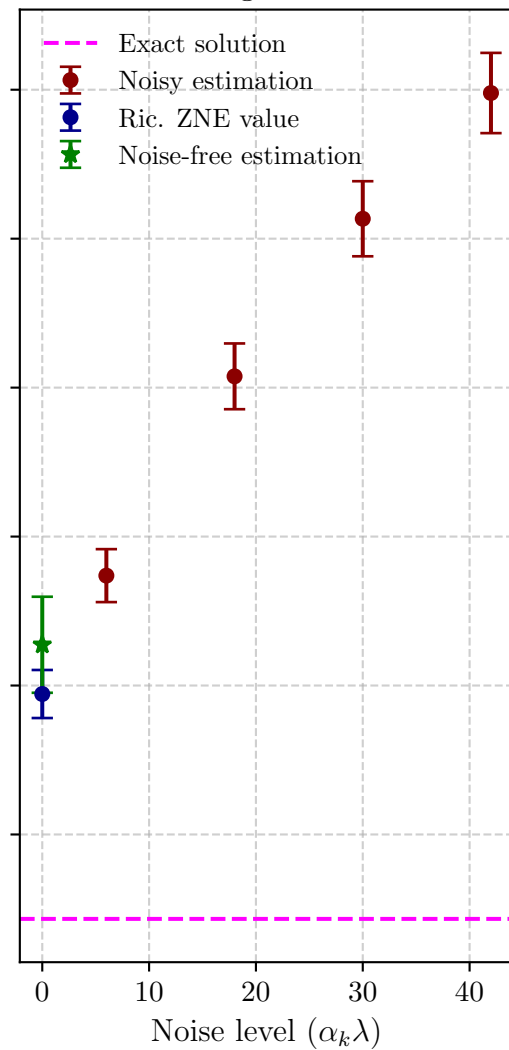
XY ansatz ric-4



Ising ansatz ric-4



Heisenberg ansatz ric-4



XY VQE CONFIGURATION

Parameter	Value
run	vqe
nqubits	7
state	dmatrix
output.file_name_prefix	xy_noisefree_time_evo
output.draw.status	True
output.draw.fig_dpi	100
output.draw.type	png
observable.def	ising
observable.coefficients.cn	[0.5, 0.5, 0.5, 0.5, 0.5, 0.5]
observable.coefficients.bn	[1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0]
observable.coefficients.r	1
ansatz.layer	30
ansatz.gateset	1
ansatz.ugate.type	xy-iss
ansatz.ugate.coefficients.cn	[0.5, 0.5, 0.5, 0.5, 0.5, 0.5]
ansatz.ugate.coefficients.bn	[0, 0, 0, 0, 0, 0, 0]
ansatz.ugate.coefficients.r	0
ansatz.ugate.time.min	0.0
ansatz.ugate.time.max	10.0
vqe.iteration	10
vqe.optimization.status	True
vqe.optimization.algorithm	SLSQP
vqe.optimization.constraint	True
init_param.value	random
noise_profile.status	True
noise_profile.type	dephasing
noise_profile.noise_prob	[0.001, 0.001, 0.0, 0.0]
noise_profile.noise_on_init_param.status	False
noise_profile.noise_on_init_param.value	0
redundant.identity_factors	[[0, 0, 0, 0], [1, 1, 1, 1]]
zne.method	richardson
zne.degree	1
zne.sampling	default
zne.data_points	

ISING VQE CONFIGURATION

Parameter	Value
run	vqe
nqubits	7
state	dmatrix
output.file_name_prefix	ising_noisefree_time_evo
output.draw.status	True
output.draw.fig_dpi	100
output.draw.type	png
observable.def	ising
observable.coefficients.cn	[0.5, 0.5, 0.5, 0.5, 0.5, 0.5]
observable.coefficients.bn	[1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0]
observable.coefficients.r	1
ansatz.layer	30
ansatz.gateset	1
ansatz.ugate.type	ising
ansatz.ugate.coefficients.cn	[0.5, 0.5, 0.5, 0.5, 0.5, 0.5]
ansatz.ugate.coefficients.bn	[0, 0, 0, 0, 0, 0, 0]
ansatz.ugate.coefficients.r	0
ansatz.ugate.time.min	0.0
ansatz.ugate.time.max	10.0
vqe.iteration	10
vqe.optimization.status	True
vqe.optimization.algorithm	SLSQP
vqe.optimization.constraint	True
init_param.value	random
noise_profile.status	True
noise_profile.type	dephasing
noise_profile.noise_prob	[0.001, 0.001, 0.0, 0.0]
noise_profile.noise_on_init_param.status	False
noise_profile.noise_on_init_param.value	0
redundant.identity_factors	[[0, 0, 0, 0], [1, 1, 1, 1]]
zne.method	richardson
zne.degree	1
zne.sampling	default
zne.data_points	

HEISENBERG VQE CONFIGURATION

Parameter	Value
run	vqe
nqubits	7
state	dmatrix
output.file_name_prefix	heisenberg_noise_free_time_evo
output.draw.status	True
output.draw.fig_dpi	100
output.draw.type	png
observable.def	ising
observable.coefficients.cn	[0.5, 0.5, 0.5, 0.5, 0.5, 0.5]
observable.coefficients.bn	[1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0]
observable.coefficients.r	1
ansatz.layer	30
ansatz.gateset	1
ansatz.ugate.type	heisenberg
ansatz.ugate.coefficients.cn	[0.5, 0.5, 0.5, 0.5, 0.5, 0.5]
ansatz.ugate.coefficients.bn	[0, 0, 0, 0, 0, 0, 0]
ansatz.ugate.coefficients.r	0
ansatz.ugate.time.min	0.0
ansatz.ugate.time.max	10.0
vqe.iteration	10
vqe.optimization.status	True
vqe.optimization.algorithm	SLSQP
vqe.optimization.constraint	True
init_param.value	random
noise_profile.status	True
noise_profile.type	dephasing
noise_profile.noise_prob	[0.001, 0.001, 0.0, 0.0]
noise_profile.noise_on_init_param.status	False
noise_profile.noise_on_init_param.value	0
redundant.identity_factors	[[0, 0, 0, 0], [1, 1, 1, 1]]
zne.method	richardson
zne.degree	1
zne.sampling	default
zne.data_points	